LAB 14

Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve

Item	Detail
Topic	Topic 3: Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals
Pipeline stage	Pairs / stat-arb
Duration	25 min
Simulator page	http://localhost:8000/#/research/pairs
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

Lion City Capital's equity desk wants a sleeve whose returns do not depend on the direction of the market. The head of trading asks for the classic statistical-arbitrage hedge as a pipeline stage: hold one stock long and an economically linked stock short in hedge-ratio proportion, trade the spread when it stretches, and output per-leg target weights. Candidates: KO/PEP, JPM/BAC and XLF/JPM.

Objective

Screen pairs for cointegration and backtest a market-neutral KO/PEP spread.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 14. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - you will use it in the debrief and in the later labs.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 14: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

Step 1. Screen the pairs

Research > Pairs screen (formation to 2022-12-31).

Open: http://localhost:8000/#/research/pairs (Research > Pairs screen)

Check: High correlation does not imply cointegration - compare the corr and ADF columns.

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Step 2. Backtest KO/PEP

Backtest the pairs strategy on KO, PEP (both legs, short leg allowed, borrow cost charged).

Open: http://localhost:8000/#/backtest?strategy=pairs_cointegration&symbols=KO,PEP (Backtest KO/PEP pairs)

Check: Beta to SPY near 0 in 'All metrics'; the trade list has long and short legs.

Expected in the app: An Engle-Granger screen and a pairs backtest with beta near zero.

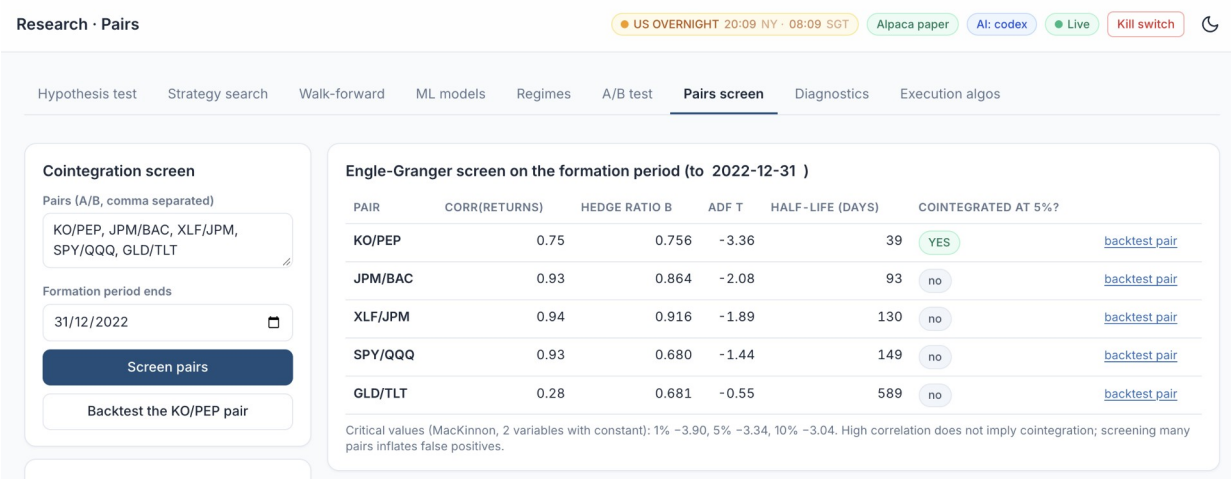


Figure: Lab 14 in the Trading Simulator.

Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs/Prompt_Library.html and .pdf.

A hedged pairs trade - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: screen KO/PEP, XOM/CVX, V/MA and HD/LOW for cointegration (Engle-Granger) on 2018-2022 and estimate each hedge ratio by OLS. Then backtest a market-neutral KO/PEP spread strategy on 2023-2025: 60-day z-score of the spread, enter when |z| > 2, exit when |z| < 0.5, long one leg and short the other, 0.05% cost per leg. Print the screen table, the trades, CAGR, Sharpe, maximum drawdown and the strategy's beta to SPY.

Open in Claude | Open in ChatGPT | Open Google Colab

Debrief

Pairs trading is a hedging algorithm: the short leg cancels most of the market exposure of the long leg, which is why the beta to SPY is near zero. The statistical model is cointegration, not correlation. A pair that is cointegrated in one period can stop reverting in the next, so professional desks re-test cointegration on a rolling basis and size each pair small.

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